higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS TECHNOLOGY
NQF LEVEL 2

(6050342)

27 February 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK. Correct the statement if it is FALSE.

1.1 The inside diameter determines how thick the material of a pipe is.

1.2 A piston pump traps fluid in its closed casing and discharges a smooth flow.

1.3 By-products are stored before being packaged, distributed or used.

1.4 A pressure filter is one of the main filters used in treating plants.

1.5 An adsorbent is a substance that adheres to a surface.

(5 × 2) **[10]**

QUESTION 2

2.1 List FIVE things that single PFDs indicate. (5)

2.2 Describe the function of lines on PFDs. (2)

2.3 Explain how equipment are represented on PFDs. (2)

2.4 Name ONE type of line that indicates piping on PFDs. (1)

2.5 Describe the main purpose of piping and fittings as used in a process environment. (2)

2.6 Piping and fittings can be manufactured from many different materials depending on the type of fluid being transported as well as the pressure and temperature of the system.

List FIVE types of piping and fitting materials used in industries. (5)

2.7 Name a type of fitting that joins two sections of a pipe. (1)

[18]

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QUESTION 3

3.1 Explain each of the following terms:

3.1.1 Sedimentation

3.1.2 Scum

3.1.3 Disinfection

(3 × 2) (6)

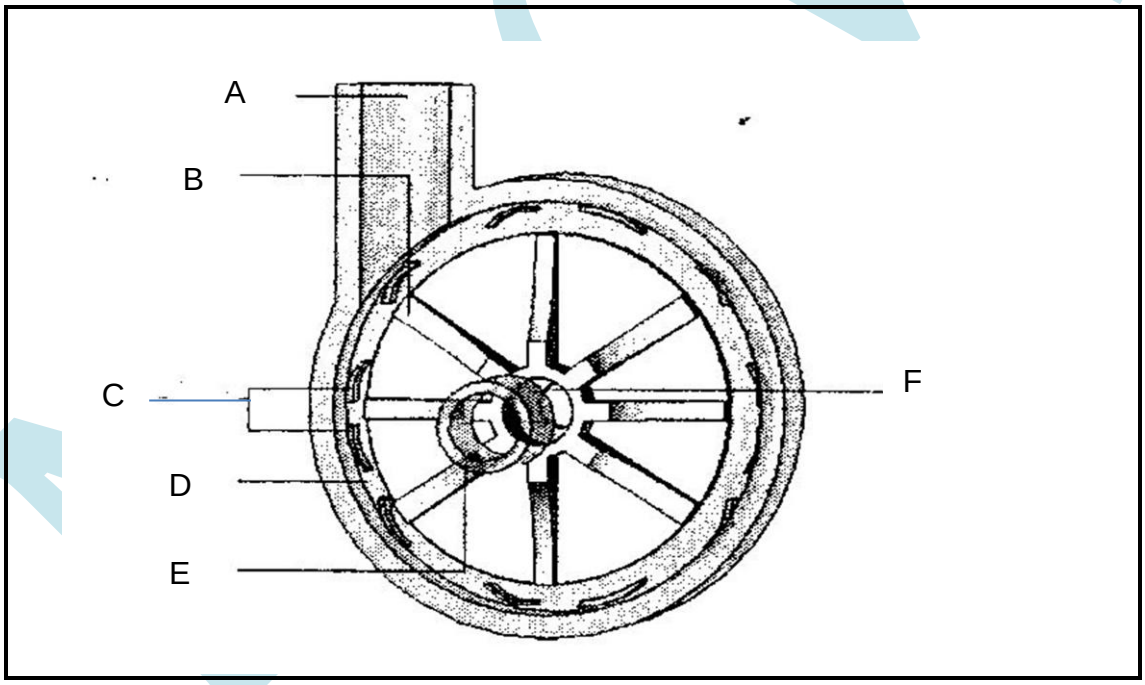
3.2 Name TWO gases that are produced in an anaerobic digester. (2)

3.3 Vibrating screens consist of a distribution panel that functions as a screen and a second screen underneath the distribution panel.

What is the common name for these screens? (2)

3.4 List FOUR applications of a piston pump. (4)

3.5 The diagram below shows the main components of a centrifugal compressor.



Label the components of the centrifugal compressor by writing only the answer next to the letter (A–F) in the ANSWER BOOK. (6 × 1) (6)

3.6 Describe the function of each of the following components of a reciprocating compressor:

3.6.1 Cylinder

3.6.2 Piston

(2 × 2) (4)

3.7 Describe the uses of scales in process operations. (4)

[28]

QUESTION 4

4.1 Products must be stored in a proper warehouse or storage area to maintain quality.

Explain EIGHT guidelines to help maintain the quality of stored products. (8)

4.2 An intermediate bulk container (IBC) is a small container with a volume of between a few 100 ℓ and 3 000 ℓ.

4.2.1 Name the type of material used to manufacture IBCs. (2)

4.2.2 What is the use of IBCs? (2)

4.3 Why is it important to store gases properly and safely? (2)

4.4 Give FOUR reasons why tanks should be coated. (4)

[18]

QUESTION 5

5.1 Explain each of the following terms:

5.1.1 Filtration

5.1.2 Counter-current decantation

5.1.3 Centrifugation

5.1.4 Cyclone process

5.1.5 Crystallisation

(5 × 2) (10)

5.2 Give FOUR uses of activated carbon as it relates to water purification. (4)

5.3 Explain how retention time affects the sedimentation process. (4)

5.4 Give TWO uses of centrifuges in the separation process. (2)

5.5 Describe the function of each of the following components of a vertical basket centrifuge:

5.5.1 Plough

5.5.2 Spray nozzle

(2 × 2) (4)

5.6 What is the main purpose of high- and medium-density cleaners? (2)

[26]

TOTAL: 100

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